

INFORMATION GATHERING TOOLS

INTRODUCTION:

Information gathering is a crucial phase in cybersecurity, penetration testing, and threat intelligence. Open-source intelligence (OSINT) tools help security professionals collect, analyse, and visualize data about a target.

This report explores six powerful information-gathering tools:

1. **Maltego**
2. **OSINT Framework**
3. **Recon FTW**
4. **Recon-ng**
5. **SpiderFoot**
6. **Aquatone.**

1. MALTEGO:

1. Tool Description:

Maltego is a powerful OSINT and forensics tool used for link analysis and data mining. It helps visualize relationships between entities like domains, IPs, emails, and social media profiles.

2. Advantages/Features/Uses:

- Graph-Based Link Analysis: Visualizes relationships between entities (domains, IPs, emails, social media).
- Transforms (Modules):
 - DNS Lookup (Domain - IP, MX records).
 - WHOIS Data Extraction (Registrant details).
 - Social Media Profiling (LinkedIn, Twitter, etc.).
 - Shodan & VirusTotal Integration (Exposes open ports, vulnerabilities).
- Customizable Workflows: Automates repetitive OSINT tasks.
- Use Cases:

- o Threat Intelligence (Tracking threat actors).
- o Corporate Espionage (Mapping organizational structures).
- o Phishing Investigations (Finding connected domains).

3. Installation Steps:

1. Download Maltego from <https://www.maltego.com/>
2. Register for a free or paid account.
3. Install the application (Windows, Linux, or macOS).
4. Launch Maltego and log in with your credentials.

<https://www.maltego.com/images/uploads/2021/07/maltego-ce-download.png>

2. OSINT FRAMEWORK:

1. Tool Description:

OSINT Framework is a web-based collection of OSINT tools organized for efficient reconnaissance. It provides categorized resources for gathering information from public sources.

2. Advantages/Features/Uses:

- Web-Based Tool Aggregator: No installation needed.
- Categories:
 - o Username Search (CheckUsernames, Sherlock).
 - o Domain/IP Tools (WHOIS, DNSDumpster).
 - o Metadata Extraction (ExifTool, FOCA).
 - o Dark Web Search (OnionScan, TorBot).
- Use Cases:
 - o Quick Reconnaissance (Before deep scanning).
 - o Password Recovery (Email/username associations).
 - o Competitor Analysis (Publicly available data).

Example Workflow:

1. Find Email - Use Hunter.io or EmailHippo.
2. Check Breaches - HaveIBeenPwned.
3. Social Media - WhatsMyName.

3. Installation Steps:

1. Visit <https://osintframework.com/>
2. Browse categories (e.g., Usernames, Domain Names, Metadata).

3. RECON FTW (Recon For The Win):

1. Tool Description:

Recon FTW automates OSINT reconnaissance by combining multiple tools like Sublist3r, Amass, and Aquatone for comprehensive target analysis.

2. Advantages/Features/Uses:

- Automated Subdomain Discovery:
 - Sublist3r, Amass, AssetFinder.
- Port Scanning:
 - Naabu, Masscan (Fast TCP/UDP scans).
- Screenshot & Tech Stack Detection:
 - Aquatone, WebTech.
- Report Generation:
 - HTML/PDF summaries.
- Use Cases:
 - Bug Bounty Hunting (Subdomain takeovers).
 - Penetration Testing (Asset discovery).

3. Installation Steps:

1. Clone the repository : `git clone https://github.com/6mile/ReconFTW.git`

`cd ReconFTW`

2. Run the installer : `chmod +x install.sh`

`./install.sh`

3. Execute the tool : `./reconftw.sh -d example.com`

<https://github.com/6mile/ReconFTW/raw/main/images/reconftw.png>

4. RECON-NG:

1. Tool Description:

Recon-Ng is a Python-based OSINT framework for web reconnaissance, similar to Metasploit but for information gathering.

2. Advantages/Features/Uses:

- Metasploit-Like Interface: Modular workflow.
- Key Modules:
 - `whois_poc` (WHOIS lookup).
 - `hunter_io` (Email enumeration).
 - `github_miner` (GitHub leaks).
 - `shodan_host` (Shodan scans).
- Workspace Management:
 - Save/load investigations.
- Use Cases:
 - Email Harvesting (Phishing campaigns).
 - GitHub Recon (Exposed API keys).

3. Installation Steps:

1. Install via apt : `sudo apt install recon-ng`
2. Launch Recon-Ng : `recon-ng`
3. Load modules (e.g., `whois_poc`) : `marketplace install whois_poc`

<https://www.kali.org/tools/recon-ng/images/recon-ng-logo.svg>

5. SPIDERFOOT:

1. Tool Description:

SpiderFoot is an automated OSINT tool that scans IPs, domains, and emails to gather threat intelligence.

2. Advantages/Features/Uses:

- 100+ Data Sources:
 - Passive DNS (SecurityTrails).
 - Breach Databases (HaveIBeenPwned).
 - Geolocation (MaxMind).
- Automated Scans:
 - IP ↔ Associated Domains.
 - Email ↔ Social Media Profiles.
- Use Cases:
 - Threat Hunting (Tracking malicious IPs).
 - Attack Surface Mapping (Exposed assets).

3. Installation Steps:

1. Install via pip : `pip install spiderfoot`
2. Launch the web interface : `spiderfoot -l 127.0.0.1:5001`
3. Access via browser : <http://127.0.0.1:5001>

6. AQUATONE:

1. Tool Description:

Aquatone is a tool for visual reconnaissance of websites, providing screenshots and service analysis.

2. Advantages/Features/Uses:

- Website Screenshots: Headless Chrome rendering.
- Technology Detection:
 - Wappalyzer Integration (CMS, frameworks).
- Port Scanning:
 - HTTP/HTTPS services.
- Use Cases:
 - Bug Bounty (Visual recon for hidden endpoints).
 - Phishing Detection (Clone site identification).

3. Installation Steps:

1. Install dependencies : `sudo apt install chromium`
2. Download Aquatone : `wget https://github.com/michenriksen/aquatone/releases/download/v1.7.0/aquatone_linux_amd64_1.7.0.zip`
`unzip aquatone_linux_amd64_1.7.0.zip`
3. Run Aquatone : `cat subdomains.txt | ./aquatone`

CONCLUSION:

Each of these tools serves a unique purpose in the OSINT (Open-Source Intelligence) and cybersecurity reconnaissance process. Here's a quick guide on when to use which tool:

- For Link Analysis & Visualization □ Maltego (Best for mapping relationships between entities).

- For Quick Web-Based Recon □ OSINT Framework (No installation, great for initial checks).
- For Automated Subdomain & Asset Discovery □ Recon FTW (Combines multiple tools for efficiency).
- For Modular & Scriptable Recon □ Recon-Ng (Metasploit-style OSINT framework).
- For Deep Threat Intelligence & Data Correlation □ SpiderFoot (Scans 100+ sources for hidden risks).
- For Website Screenshots & Visual Recon □ Aquatone (Ideal for bug bounty hunters).
- **Penetration Testers & Red Teams** □ Use Recon-Ng + Maltego for structured intelligence.
- **Bug Bounty Hunters** □ Aquatone + Recon FTW for fast target mapping.
- **Threat Analysts** □ SpiderFoot + OSINT Framework for tracking threat actors.